

SENTINEL Protocol

Multi-Agent Sepsis Detection — Project Brief

Prepared for: Business & Technical Review | Date: July 2026 | Status: Live Demo Ready

1. Executive Summary

SENTINEL is a fully-working, AI-native system that detects **sepsis early** — before a patient collapses — by running a deliberating **council of four specialist AI agents** over a patient's continuous vital-signs stream. It delivers a color-coded recommendation to the clinician, keeps the human in command, and records every decision for governance.

This is not a mock: the data pipeline, agents, deliberation, alerts, and human-in-the-loop responses are all real. The only simulated element is the physical hardware (a wearable patch and point-of-care blood tester), replaced by a deterministic device simulator that streams realistic vitals for five patient test cases.

Why it matters: Sepsis is a leading cause of in-hospital death and is time-critical — every hour of delayed antibiotics increases mortality. Standard monitoring reacts late; SENTINEL anticipates, explains, and acts, while reliably ruling out look-alike conditions to avoid unnecessary antibiotics.

2. Business Value at a Glance

The proposition in six points:

Outcome	What it means
Detect earlier	Catches sepsis ~15–45 minutes before standard monitoring triggers, within the critical 1-hour treatment window.
Avoid false alarms	Correctly keeps non-sepsis look-alikes (pancreatitis, post-op stress) at safe / green — no unnecessary antibiotics.
Explainable by design	Each alert carries per-agent reasoning, a cross-examination transcript, confidence, and a recommended action — not a black box.
Clinician in command	SENTINEL recommends; the human accepts, modifies, or rejects. Alerts work even when the AI is offline.
Governance-ready	Every alert, clinician response, and AI call is written to an immutable audit trail for compliance and review.
Carbon-aware AI	A budget controller calls the large AI model only on genuinely ambiguous cases; 80% of inferences run zero-LLM and free.

3. How It Works — End to End

Data flows in one direction, from bedside to clinician to record:

1. Step 1 — Bedside: A continuous wearable patch plus point-of-care blood tests stream heart rate, breathing rate, blood pressure, temperature, oxygen saturation, and lab values every second.
2. Step 2 — Real-time engine: A rolling 15-minute feature store computes trends, rates of change, shock index, and clinical SOFA / qSOFA organ-dysfunction scores from the raw stream.
3. Step 3 — The AI council: Four independent specialist agents each assess the patient from a different angle (see Section 4).
4. Step 4 — Deliberation: The orchestrator runs a three-stage deliberation — independent assessment, cross-examination, then synthesis into a weighted consensus.
5. Step 5 — Tiered alert: A color-coded recommendation is issued: GREEN (safe), YELLOW (review), or RED (act now).
6. Step 6 — Human-in-the-loop: The clinician accepts, modifies, or rejects the recommendation. SENTINEL recommends; the human decides.
7. Step 7 — Outcome & governance: Patient outcome is tracked (with a with-vs-without-SENTINEL counterfactual) and every event is written to the audit trail.

Key safety property: the alert trigger is always driven by structured clinical data; the AI only writes the explanation. Alerts therefore fire correctly even if the AI is slow, rate-limited, or offline — the system degrades to a deterministic rule engine, never to silence.

4. The AI Council — Four Agents Plus an Orchestrator

SENTINEL is not a single black-box model. It is a council of independent specialists that deliberate. This structure is what makes the output both more accurate and explainable.

Agent	Role	Engine
SOFA Tracker	Tracks organ-dysfunction score (Sepsis-3 SOFA) and how fast it is worsening versus the patient's own baseline.	Clinical rules
Differential	Rules out sepsis look-alikes (pancreatitis, post-op stress) so sepsis is not over-called.	Clinical rules
Predictor	Machine-learning model that estimates the probability of sepsis from the trend features.	Trained ML (XGBoost)
Narrative Analyst	Reads free-text nursing notes for concern signals that the numbers alone might miss.	Trained ML (TF-IDF + LogReg)
Carbon Budget Controller	Selects the AI model tier based on case ambiguity — small model, large model, or no model at all — to minimize cost and carbon.	Rule thresholds
Orchestrator	Runs the 3-stage deliberation, computes	Grok synthesis +

Agent	Role	Engine
	weighted consensus, assigns the alert tier, and assembles the alert payload.	deterministic tier

4.1 The 3-Stage Deliberation

- Stage 1 — Independent: each agent scores the patient on its own (the fast, zero-LLM path that handles roughly 80% of inferences).
- Stage 2 — Cross-examination: agents challenge each other's reasoning; disagreements are surfaced explicitly.
- Stage 3 — Synthesis: a weighted consensus is reached and mapped to an alert tier, with a recommended action and an uncertainty flag.

5. The Live Demo — Five Patients, One Screen

All five patients stream simultaneously on a single dashboard. Two real sepsis cases escalate correctly and are caught early; two mimics stay safely green; one healthy control is monitored silently. This is the proof that the system is both sensitive and specific.

Patient	Ward	Case	Expected behaviour
Karen Whitfield	ER	Sepsis (headline)	GREEN → YELLOW (~t=30) → RED (~t=44). Caught before collapse.
Anya Petrova	ICU	Septic shock	RED early, then de-escalates to YELLOW as she recovers.
Maria Sanchez	ER	Pancreatitis (mimic)	Stays GREEN — the Differential agent rules sepsis out.
Robert Hayes	Surgical	Post-op stress (mimic)	Stays GREEN — distinguished from sepsis.
James Okafor	General	Healthy control	Stays GREEN — silent monitoring throughout.

The clinician can accept an alert (Karen's outcome flips to 'stable — intervention timely'), open any patient for live vitals charts, POCT labs, SOFA organ breakdown, agent reasoning cards, and the audit / governance panel.

6. Technology Stack

A deliberately lean, production-shaped stack — no heavyweight infrastructure required to run the demo.

Layer	Choice
Language	Python 3.10
Backend	FastAPI + Uvicorn (ASGI), async-native
Real-time transport	Native WebSocket with pub/sub broadcast
AI / deliberation	xAI Grok (grok-4, grok-4-fast) via OpenAI-compatible SDK; deterministic local fallback
ML scoring	XGBoost + scikit-learn (TF-IDF + LogisticRegression), serialized with joblib
Clinical logic	Sepsis-3 SOFA / qSOFA rule engine; mimic rule-out; hysteresis tier assignment
Data layer	In-memory state + append-only JSONL audit trail (no database)
Frontend	Vanilla HTML/CSS/JS single page, offline-vendored charting
Validation / config	Pydantic v2; .env + python-dotenv

6.1 Trained Models (real, not faked)

- Predictor: XGBoost sepsis-risk classifier trained on a synthetic cohort replayed from the five patient profiles (7,200 samples). Holdout AUC 1.00, sepsis recall 0.99.
- Narrative: TF-IDF + LogisticRegression nursing-note concern classifier, trained on augmented notes.
- Both load lazily with automatic rule fallback if a model file is missing, so the demo never breaks. Retrain anytime via a single command (~10 seconds, deterministic).

7. Integration Surface (for technical reviewers)

The backend exposes a small, clean REST + WebSocket API. A hospital pilot would replace the device simulator with real EHR / HL7 feeds at the same boundary.

Endpoint	Purpose
GET /api/state	Full live snapshot of all patients
POST /api/device/{id}/connect	Connect a device patch for a patient
POST /api/speed	Set demo speed (0.5×–4×)
POST /api/alert/{id}/respond	Clinician HITL response: ACCEPT / MODIFY / REJECT
GET /api/audit	Append-only audit entries
GET /api/metrics	Batch KPIs: sensitivity, specificity, time-to-alert, equity, drift
WS /ws	Live state stream to the dashboard

7.1 Batch Metrics & Safety Monitoring

Beyond the live demo, a batch runner computes the brief's headline measures over a synthetic cohort:

- KPIs: sensitivity, specificity, median time-to-alert, override rate, mortality reduction vs no-SENTINEL, net carbon saved per life.
- Equity: AUROC disaggregated by sex and age bucket (SDG 10 fairness checkpoint).
- Drift: AUROC over chronological windows with SPC control limits and a qSOFA auto-fallback flag.

8. Honest Limitations & Next Steps

Stated plainly, as a matter of engineering integrity:

- SOFA uses practical proxies (SpO_2 for $\text{PaO}_2/\text{FiO}_2$, MAP without vasopressors) from demo data. It follows Sepsis-3 structure but is not a validated clinical device.
- Vitals are scripted and interpolated with reproducible noise so every demo run is identical and predictable.
- No EHR / HL7 integration yet — this is the #1 technical risk for a hospital pilot and the primary next step.
- No database; state is in-memory and the audit log is a local file. A pilot would move these to a persisted, audited store.

8.1 Proposed Roadmap

1. Phase 1 — Standalone demo (complete): the build described in this document.
2. Phase 2 — Hospital pilot: real EHR / HL7 ingestion, persisted state and audit, clinical validation of the SOFA proxies.
3. Phase 3 — Multi-site evaluation: equity and drift monitoring in production, model retraining pipeline, regulatory engagement.

9. Running the Demo

One command launches the full system — backend, dashboard, and device simulator:

```
cd /home/devansh/Documents/icam_friday && ./sentinel/run.sh
```

Then open <http://localhost:8000> in a browser. The system runs fully without an AI key; add an xAI API key to `.env` to enable real Grok deliberation from the agents and orchestrator.

Companion artifacts in the repository: **TECH_STACK.md** (full technical reference) and **static/workflow.html** (a visual workflow diagram for non-technical audiences).